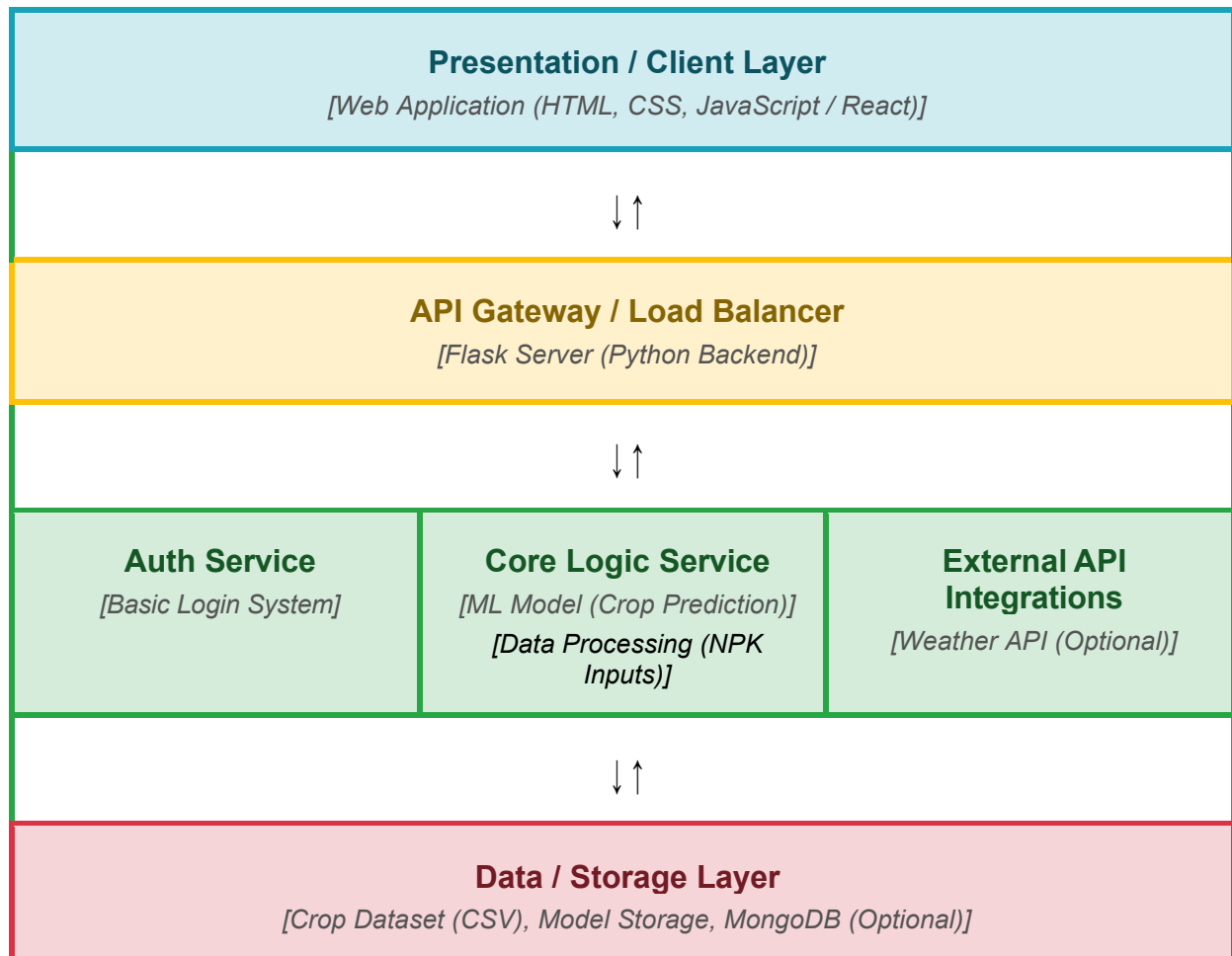


## Solution Architecture

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 9 July 2026                                                 |
| <b>Team ID</b>       | Not Assigned                                                |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                     |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name              | Description / Role in Architecture                                                     | Technologies Used                          |
|-----------------------------|----------------------------------------------------------------------------------------|--------------------------------------------|
| <i>[Presentation Layer]</i> | <i>Provides user interface for farmers to input data and view crop recommendations</i> | <i>HTML, CSS, JavaScript / React</i>       |
| <i>[API Gateway]</i>        | <i>Handles requests between frontend and backend, processes API calls</i>              | <i>Flask (Python)</i>                      |
| <i>[Microservices]</i>      | <i>Performs core logic like data processing and crop prediction using ML model</i>     | <i>Python, Scikit-learn, Pandas, NumPy</i> |
| <i>[Database]</i>           | <i>Stores dataset, trained model, and optionally user data</i>                         | <i>CSV Dataset, MongoDB (optional)</i>     |